Buổi 02

Hà Lê Hoài Trung

Nội dung

- Introduction to Cloud Computing
- Cloud computing platform

Introduction to Cloud Computing

- WHAT IS CLOUD COMPUTING ?
- Properties and Characteristics
- Benefits From Cloud

WHAT IS CLOUD COMPUTING ?

- Cloud computing is a paradigm of computing, a new way of thinking about IT industry but not any specific technology.
- Central ideas
    - Utility Computing
    - SOA - Service Oriented Architecture
    - SLA - Service Level Agreement
- Properties and characteristics
    - High scalability and elasticity
    - High availability and reliability
    - High manageability and interoperability
    - High accessibility and portability
    - High performance and optimization
- Enabling techniques
    - Hardware virtualization
    - Parallelized and distributed computing
    - Web service

Properties and Characteristics

- What Is Web Service?
- Service Oriented Architecture
- Quality Of Service
- Service Level Agreement
- Scalability & Elasticity
    - Dynamic provisioning
    - Multi-tenant design
- Availability & Reliability
    - Fault tolerance system
    - Require system resilience
    - Reliable system security

Properties and Characteristics

- System Resilience
  - Backup
  - Preparing
- System Security
  - Data Protection
  - Identity Management
  - Application Security
  - Privacy
- Manageability & Interoperability
  - System control automation
  - System state monitoring
  - Billing System

Properties and Characteristics

- Performance & Optimization
  - Parallel computing
  - Load balancing
  - Job scheduling
- Accessibility & Portability
  - Uniform access
  - Thin client

Benefits From Cloud

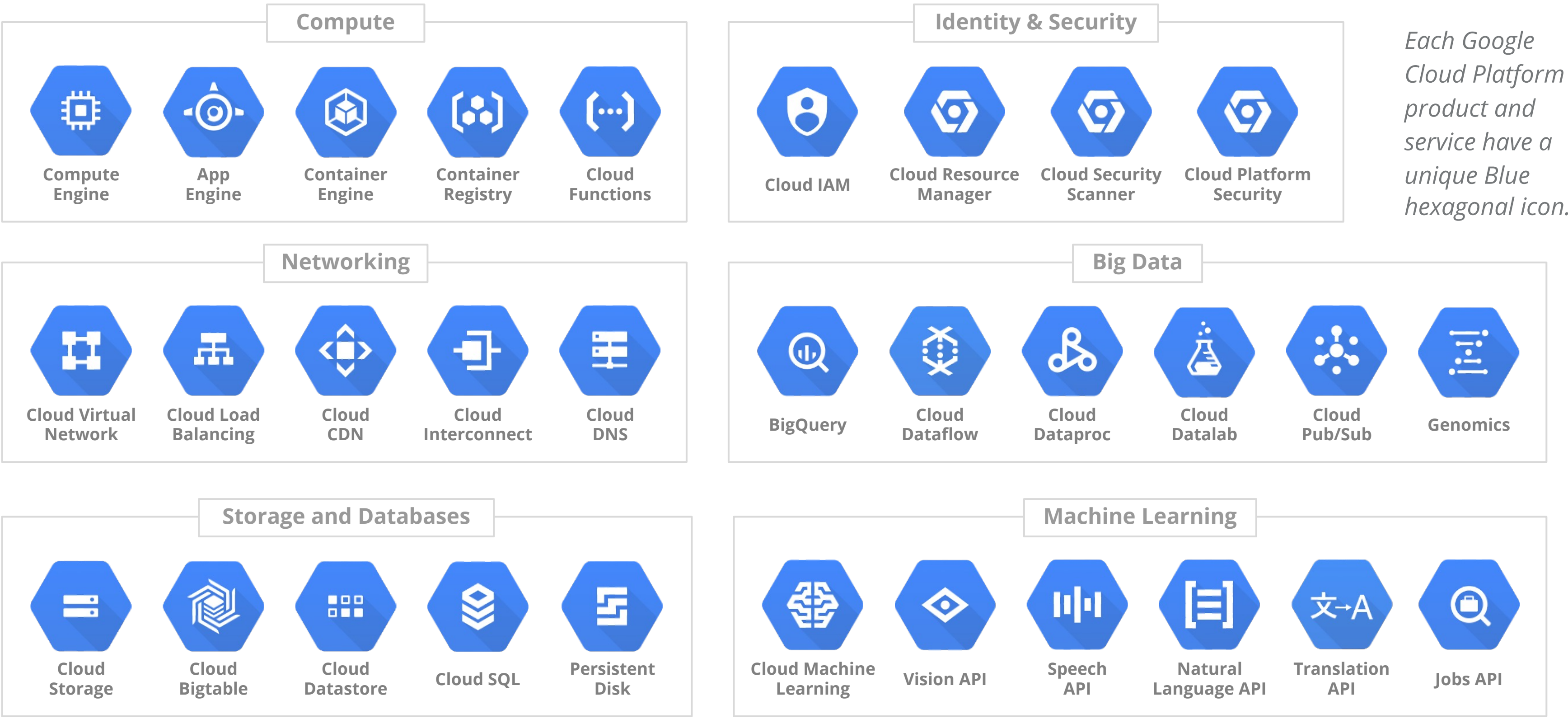
- Cloud computing brings many benefits :
  - For the market and enterprises
    - Reduce initial investment
    - Reduce capital expenditure
    - Improve industrial specialization
    - Improve resource utilization
  - For the end user and individuals
    - Reduce local computing power
    - Reduce local storage power
    - Variety of thin client devices in daily life

Cloud computing platform

- Google cloud platform
- Microsoft Azure
- Amazon Web Services

Google cloud platform

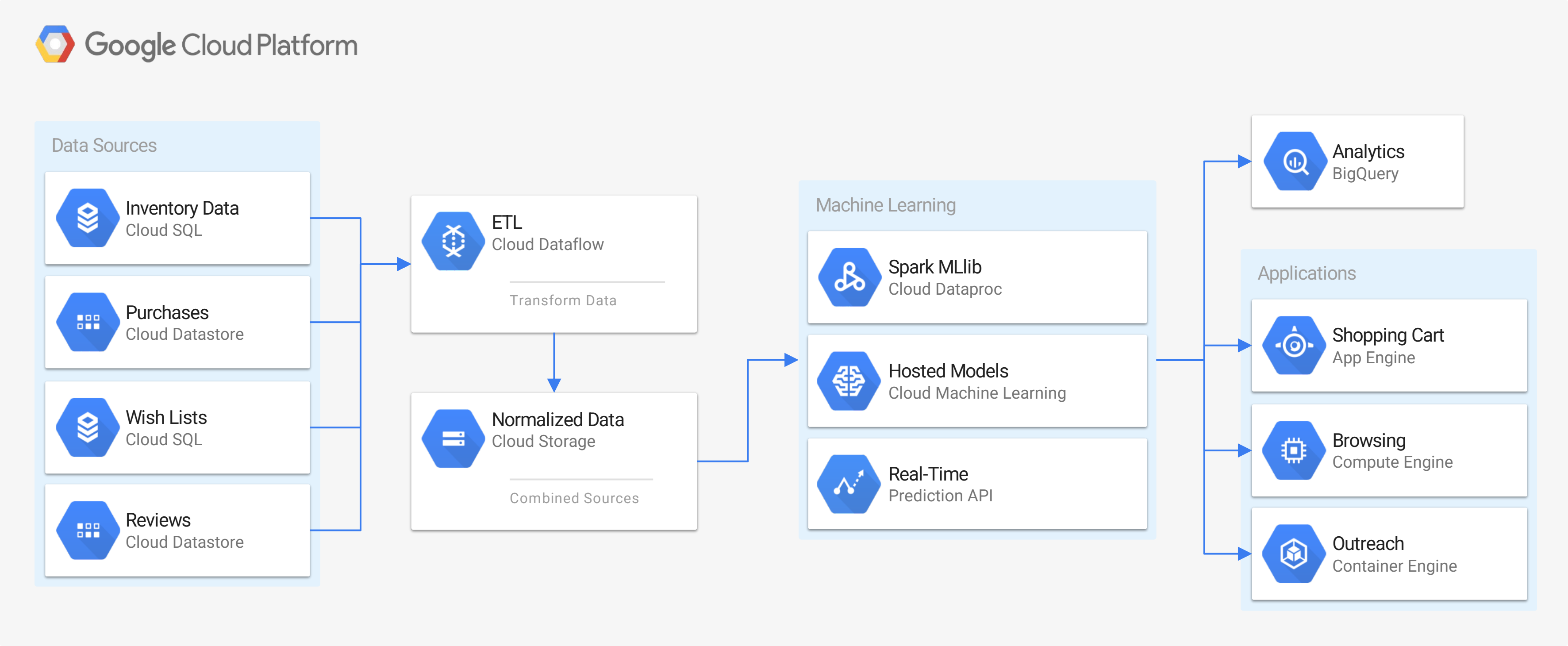
- Overview



Each Google Cloud Platform product and service have a unique Blue hexagonal icon.

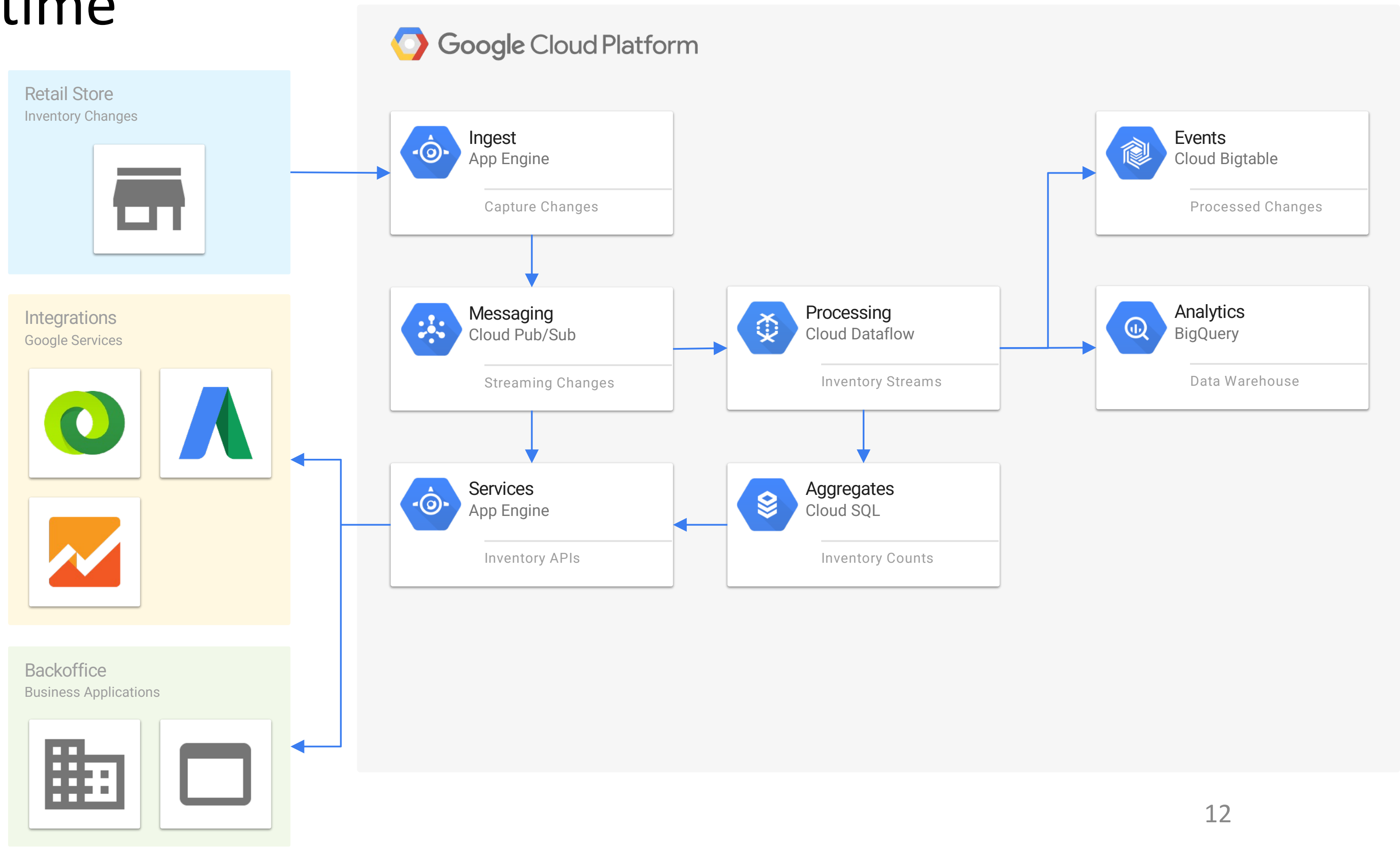
Google cloud platform

- Retail & Recommendation Engine
  - Spark Mlib
  - Cloud Dataproc
  - Cloud Dataflow
  - Cloud Datastore
  - BigQuery



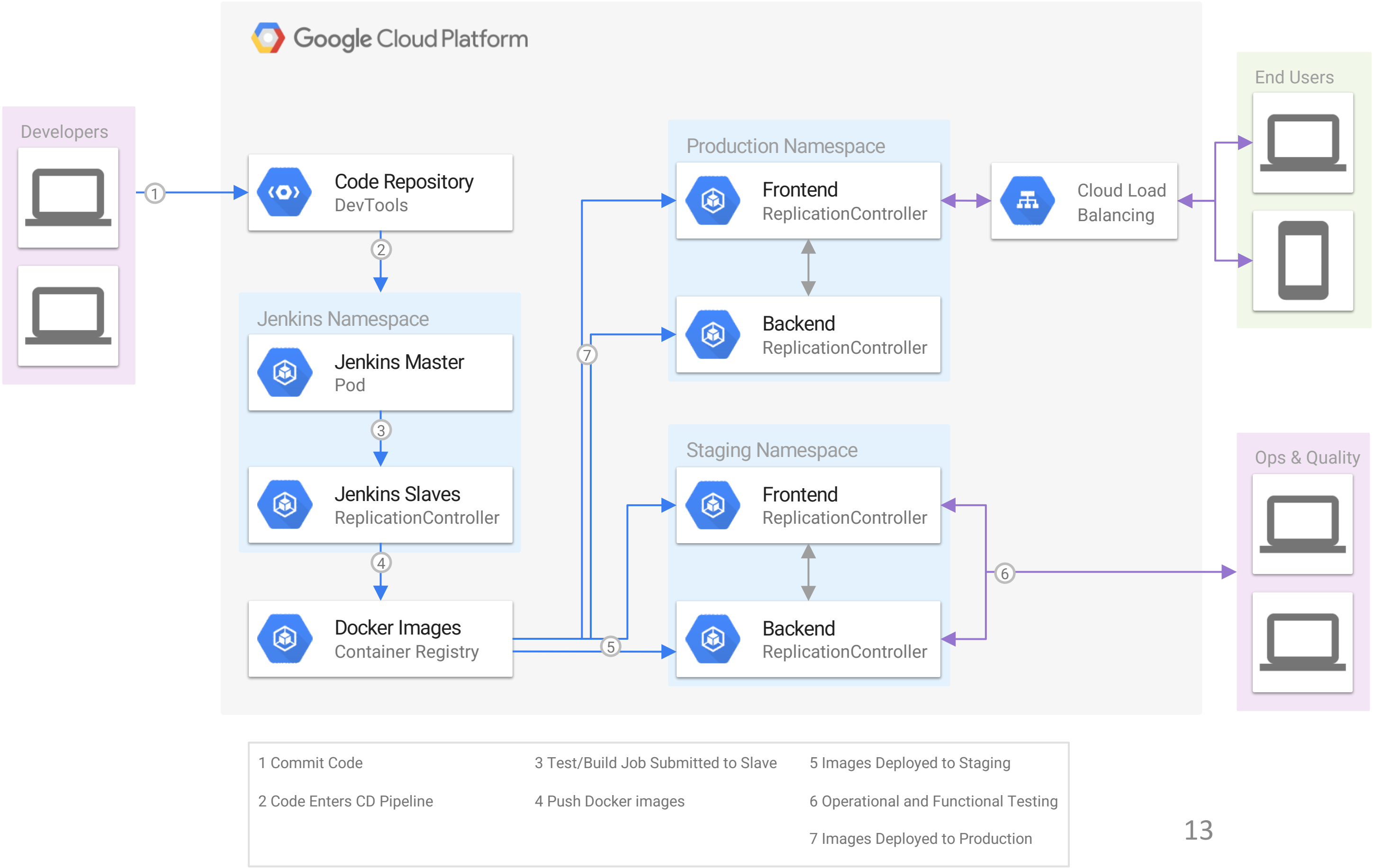
Google cloud platform

- Retail – Inventory real-time
  - Cloud Pub/Sub
  - Cloud Dataflow
  - Cloud SQL
  - BigQuery
  - BigTable

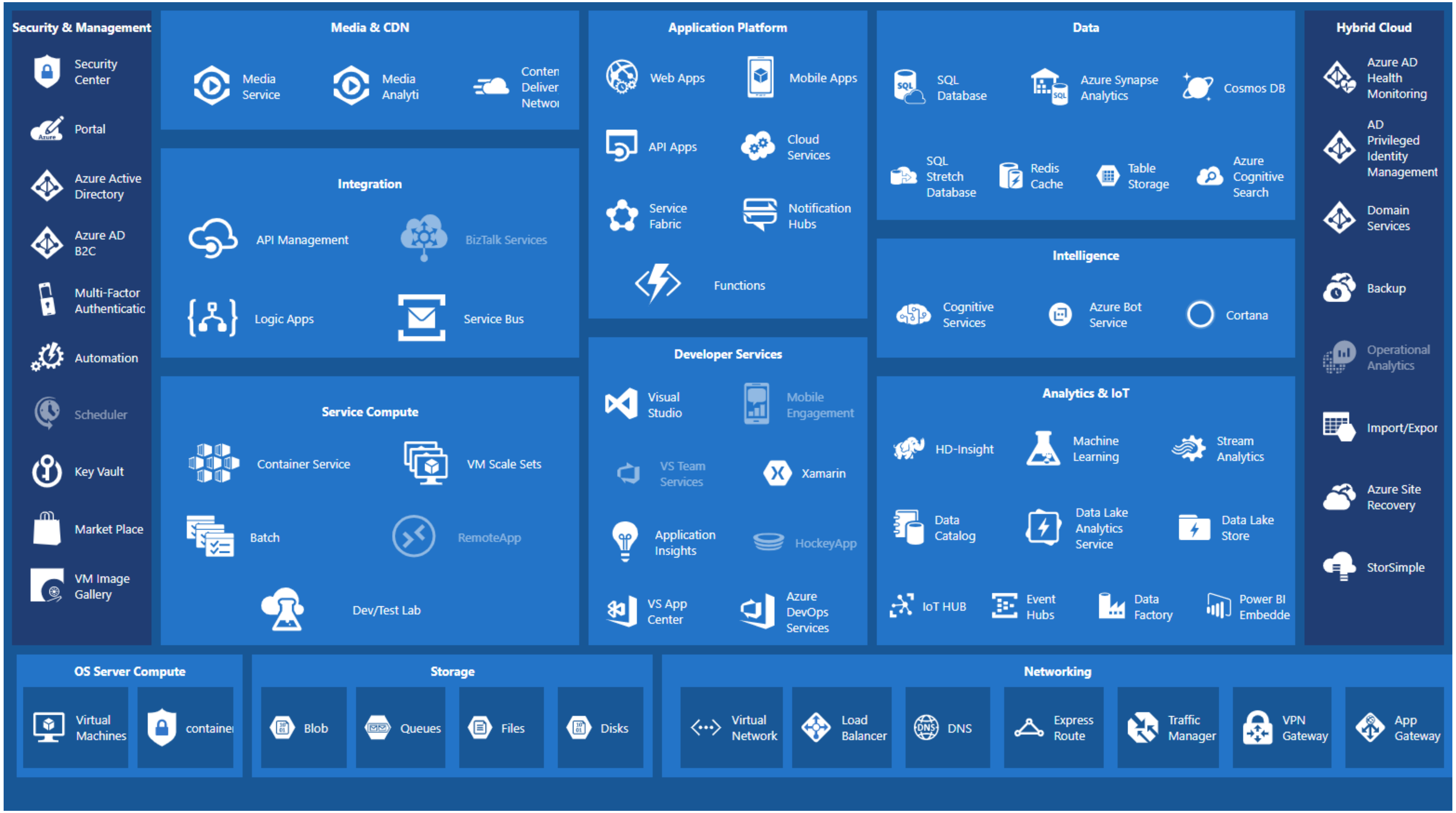


Google cloud platform

- CI/CD - Continuous Integrated / Continuous Delivery
  - Code Repository
  - Jenkins
  - Docker
  - Cloud Load Balancing
  - Spinnaker
  - Kubernetes+Locust



Microsoft Azure



Microsoft Azure

- Compute
  - Azure Function
  - Virtual Machine Scale sets
  - App Service
- Network
  - Load Balancer
  - Application Gateway
  - Content Delivery Network
- Database
  - CosmosDB
  - Azure SQL Database
  - Azure Database Migration Service
  - Azure SQL Data Warehouse
- IoT Hub
- Data Lake Analytics

Amazon Web Services

- Feature Service
  - Amazon Lightsail
  - Amazon EC2
  - Amazon simple storage service
- Database
  - Amazon RDS
  - Amazon ElastiCache
  - Amazon DynamoDB
- Analytics
  - Amazon OpenSearch service
  - Amazon Redshift
- Containers
  - Amazon Elastic Container Registry

Amazon Web Services

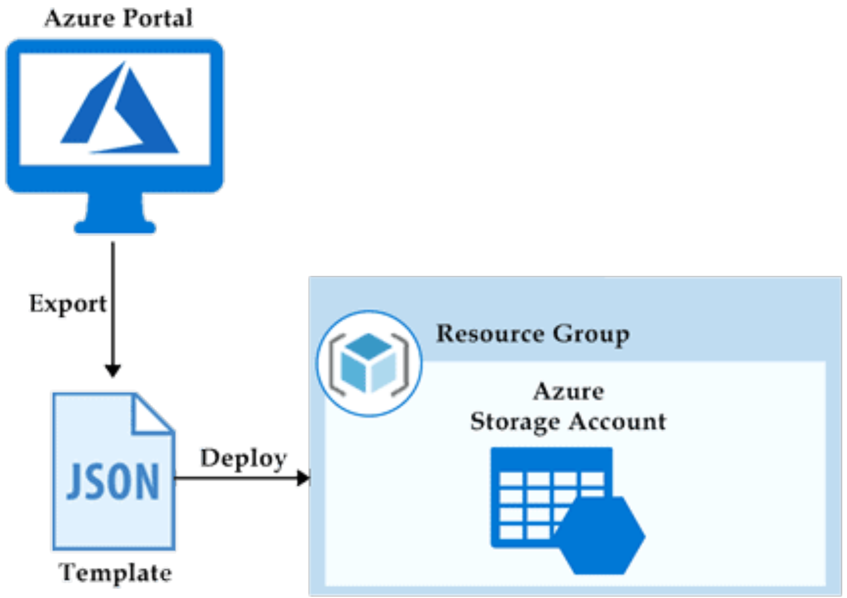
- Front-end web
  - Amazon API gateway
  - AWS App sync
  - AWS device farm
  - Amazon Loaction service
  - AWS Amplify
- IoT
  - AWS IoT Greengrass
- Machine Learning
  - Amazon SageMaker
  - Amazon Rekognition

Amazon Web Services

- Network & Content Delivery
  - Amazon CloudFront
  - Amazon API Gateway
  - Elastic Load Balancing
- Security
  - Amazon GuardDuty
  - AWS Key Management Service
- Serverless
  - AWS Lambda

Bài tập tuần 02

- WORKING WITH AZURE STORAGE USING POWERSHELL
  - Azure Resource Manager (ARM) - IaC
  - Blob, Table (Bảng) và Queue (Hàng đợi)
  - Sử dụng các lệnh azure trên môi trường power shell
  - Shared Access Signature (SAS)
  - Storage Account Keys



Bài tập tuần 02

- Managing Azure Storage Accounts
  - Azure Storage Account
  - Access Keys và Shared Access Signature
    - Dịch vụ
      - ✓ Blob
      - ✓ File
      - ✓ Queue
      - ✓ Table
    - Tài nguyên
      - ✓ Service
      - ✓ Container

Bài tập tuần 02

- WORKING WITH AZURE COSMOS DB
  - CosmosDB
  - Đặc trưng
    - Multi-model
    - Partition
    - Request Unit & throughout level
    - Global distribution
    - Consistency levels
  - Thao tác
    - Khởi tạo
    - Cấu hình: manual & auto
    - Global
  - Connect Azure Cosmos DP ABI cho MongoDB